

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a **title**, **author**, **genre**, and a **fixed number of physical copies in stock**. A registered member can borrow an available book copy and must return it within 14 days. **If the book is returned late**, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are **notified when it becomes available**. The librarian needs to **see all currently active loans**, **view all overdue loans sorted by how late they are**, and **get a report of the most-borrowed books ranked by total borrow count**.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>Book</u>		
Property Name	Data Type	Source Tag
<u>bookid</u>	<u>int</u>	<u>system generated</u>
<u>title</u>	<u>string</u>	<u>user input</u>
<u>author</u>	<u>string</u>	<u>user input</u>
<u>genre</u>	<u>string</u>	<u>user input</u>
<u>totalcopies</u>	<u>int</u>	<u>calculated</u>
<u>borrowCount</u>	<u>int</u>	<u>calculated</u>

Entity Name: <u>Member</u>		
Property Name	Data Type	Source Tag
<u>memberid</u>	<u>int</u>	<u>system generated</u>
<u>memberName</u>	<u>string</u>	<u>user input</u>
<u>phone</u>	<u>string</u>	<u>user input</u>
<u>email</u>	<u>string</u>	<u>user input</u>
<u>registrationDate</u>	<u>date</u>	<u>calculated</u>
<u>isActive</u>	<u>bool</u>	<u>default value</u>

Entity Name: <u>loan</u>		
Property Name	Data Type	Source Tag
<u>loanid</u>	<u>int</u>	<u>system generated</u>
<u>bookid</u>	<u>int</u>	<u>from list</u>
<u>memberid</u>	<u>int</u>	<u>from list</u>
<u>borrowDate</u>	<u>date</u>	<u>calculated</u>
<u>dueDate</u>	<u>date</u>	<u>calculated</u>
<u>status</u>	<u>bool</u>	<u>default value</u>
<u>fine</u>	<u>double</u>	<u>calculated</u>

Entity Name: <u>reservation</u>		
Property Name	Data Type	Source Tag
<u>reservationid</u>	<u>int</u>	<u>system generated</u>
<u>memberid</u>	<u>int</u>	<u>from list</u>
<u>bookid</u>	<u>int</u>	<u>from list</u>
<u>reservationDate</u>	<u>date</u>	<u>calculated</u>
<u>status</u>	<u>bool</u>	<u>default value</u>
<u> </u>	<u> </u>	<u> </u>
<u> </u>	<u> </u>	<u> </u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

library -> manage (books, members, loans, reservation)

1- member must first register before borrow or reserve book

2- librarian add book to the system, and each book has own info such as(title, author , genre and number of available copies)

3- member after register if borrows book the system create loan record, the available copies of book increase by 1 and the due date is set to 14 days after borrowing

4- when the member return book the loan is updated with return date, the available copies of book increase by 1 and if the member returns late the system calculates as a fine.

5- member can create reservation if there is no available book copies and if the book becomes available the system send notification to member.

6- the librarian can view active loans, overdue loans and see the most borrowed books report.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	<code>registerMember()</code>	add new mwmbner to system
2	<code>addBook()</code>	for librarian to add book with its info
3	<code>borrowBook()</code>	allow member to borrow available book and returns loan book
4	<code>ReturnBook()</code>	allow member to return book and updates number of copies
5	<code>reserveBook()</code>	allow member to reserve not available book
6	<code>calculateFine()</code>	calculate fine the member late return a book
7	<code>notifyMember()</code>	notify member if the book becomes available
8	<code>viewActiveLoans()</code>	to see all currently active loans
9	<code>viewOverdueLoans()</code>	view all overdue loans sorted by how late they are
10	<code>GetReport()</code>	get a report of the most-borrowed books ranked by total borrow count.

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
guestId	int	<u>system generated</u>
guestName	string	<u>user input</u>
guestPhone	string	<u>user input</u>
guestEmail	string	<u>user input</u>
guestNationality	string	<u>user input</u>
registrationDate	string	<u>system generated</u>

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	<u>system generated</u>
roomNumber	string	<u>user input</u>
roomType	string	<u>from list</u>
floorNumber	int	<u>from list</u>
nightlyPrice	decimal	<u>from list</u>
isAvailable	bool	<u>system generated</u>

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	<u>system generated</u>
guestId	int	<u>from list</u>
roomId	int	<u>from list</u>
checkInDate	string	<u>calculated</u>
checkOutDate	string	<u>calculated</u>
numberOfNights	int	<u>user input</u>
totalPrice	decimal	<u>calculated</u>
status	string	<u>default value</u>

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	system generated
bookingId	int	from list
guestId	int	from list
roomId	int	from list
actualCheckIn	string	system generated
actualCheckOut	string	system generated
finalAmount	decimal	calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:
 guest can bookings, and booking for one room, if the guest booked room the room become unavailable. room has different type(single, double, suite) and nightly rate a room can have one active booking. the chickinrecord links the guest, booking and room together and stores check in date. the finalamount calculated from number of nights and rooms nightly rooms. when the guest check out the room changes back to available so it can booked by another guest. if booking is canceled before check in it will be cancelled and the room will immediately made available again.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	registerGuest()	add new guest in the system
2	booking()	books room for guest and marks room unavailable
3	checkin()	make check in records when guest arrives
4	cancelBooking()	cancel booking before check in
5	checkout()	make check out record
6	calculateAmount()	calculat the total nights stayed
7	status()	change rooms availability during the guest booked or cancell or checked out

#	Service Name	What it does
8		
9		